

Members:

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Discord link: <https://discord.gg/nvdqhe8t>

<https://www.when2meet.com/?38425131-Ew2kG>

Questions for Advisor:

- Will we be supporting both AR and VR, or choosing one of the two? Is designing and building both similar to the difficulty of building just one?
- What level of resolution should our visualization include? For example, should we be able to see individual signals/bits inside components, higher level visualization of components, etc?
- How polished should the functional prototype be, and what should be the most developed areas of the project?
- What assumed level of education does the target audience have?
- What does “simplified ARM-based processor” mean? What should be excluded that would be in a full design?
- Will we be testing with the provided AR/VR devices at home or are they staying on campus/with you?
- What is the feasibility of running usability tests from sample users from our target audience?
- Is there an expanded project timeline in regards to progress?

Project description summary: We are building an XR framework/app that will be used to help students visualize common hardware and microarchitectural security issues present in computers

Immersive Hardware Security Education Framework with AR/VR Devices

Project Description

Microarchitectural attacks exploit shared processor components such as caches, pipelines, and memory systems to leak information between applications. These attacks can be difficult for students to understand because the underlying hardware behavior is largely invisible. AR/VR provides an opportunity to make these abstract concepts more intuitive by allowing students to interactively visualize processor components and observe how instructions and attacks change the microarchitectural state.

The goal of this senior design project is to develop an interactive AR/VR educational platform for teaching microarchitectural security concepts. The platform will visualize a simplified

ARM-based processor, including registers, processor front-end and back-end components, caches, and memory. Students should be able to execute selected instruction sequences, observe corresponding hardware state changes, step forward and backward through execution, and visualize how microarchitectural attacks influence shared processor resources.

The system should:

- Provide an interactive 3D visualization of major ARM processor components and their states.
- Support step-by-step execution and visualization of selected instruction sequences.
- Allow users to move backward and forward through execution states.
- Demonstrate at least one representative microarchitectural attack or side-channel scenario.
- Provide an intuitive interface suitable for students without extensive hardware-security experience.
- Support AR/VR headsets while also allowing the visualization to be mirrored to a desktop display for accessibility and users who experience motion sickness.
- Maintain sufficient performance to provide a smooth and responsive AR/VR experience.

The project will primarily use Unity to develop the interactive AR/VR environment. A simplified model of an ARMv8 processor architecture will be implemented to represent instruction execution and microarchitectural state transitions. C# will be used for application logic and visualization control, while Vulkan and/or Metal may be used through Unity for cross-platform graphics support. Development and testing will use available XR devices such as Meta Quest and Apple Vision Pro. The team will also develop example instruction sequences and microarchitectural-security demonstrations that can later be incorporated into undergraduate/graduate computer architecture and cybersecurity laboratory assignments.

Expected Project Deliverables

Expected deliverables include a functional AR/VR prototype that visualizes key ARM microarchitectural components and state changes, supports step-by-step instruction execution, and demonstrates at least one microarchitectural attack. The team will also deliver example educational scenarios/lab exercises, system documentation, source code, and a final project demonstration.

Anticipated Project Timeline

Week 20: Implement instruction execution, state-transition visualization, and interactive controls.
Week 25: Integrate and demonstrate representative microarchitectural attack scenarios.
Week 29: System testing, refinement, documentation.

Resources Provided By Client

Meta Quest 3, Meta Quest 3s, Apple Vision Pro, RayBan Smart Glasses, Pico XR headset, Apple Vision Pro, Unity, expertise on XR devices and microarchitecture attacks.

Anticipated Technical Expertise/Focus Areas

The team will utilize and develop skills in computer architecture, cybersecurity, AR/VR development, and software engineering. Relevant technical expertise may include ARM processor architecture or RISC-V/MIPS ISA, cybersecurity concepts, Unity and C# development, 3D visualization, and AR/VR interface design. nThe project will also require teamwork, testing and debugging, technical communication, and a user-centered mindset for developing an effective educational platform.

Anticipated Client Interaction

Weekly

Preferred Client Interaction Mode

In-person